1T MoE LoRA RL with veRL

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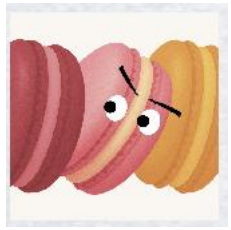
2026.01.10

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https://macaron.im/mindlab

Why "1T MoE RL"?

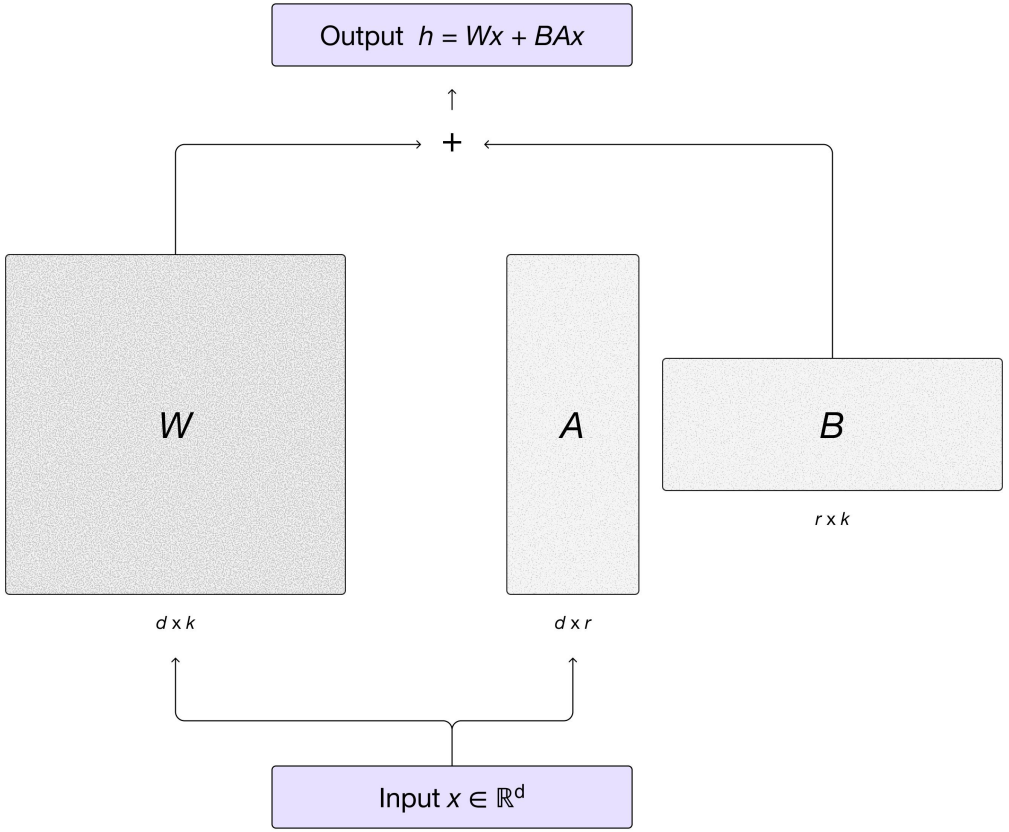
- 1T MoE open-source models (Kimi-K2[1], Ring-1T[2]) have recently emerged and reach frontier-level performance.
- RL is still required at 1T scale.
  - Some tricks work on small models, but fail on trillion-parameter models.
- Scaling changes the problem itself.

- The problem is cost.

[1] Kimi K2: Open Agentic Intelligence (Team Kimi et al, 2025)
[2] Every Step Evolves: Scaling Reinforcement Learning for Trillion-Scale Thinking Model (Team Ling et al, 2025)

Why "LoRA"?

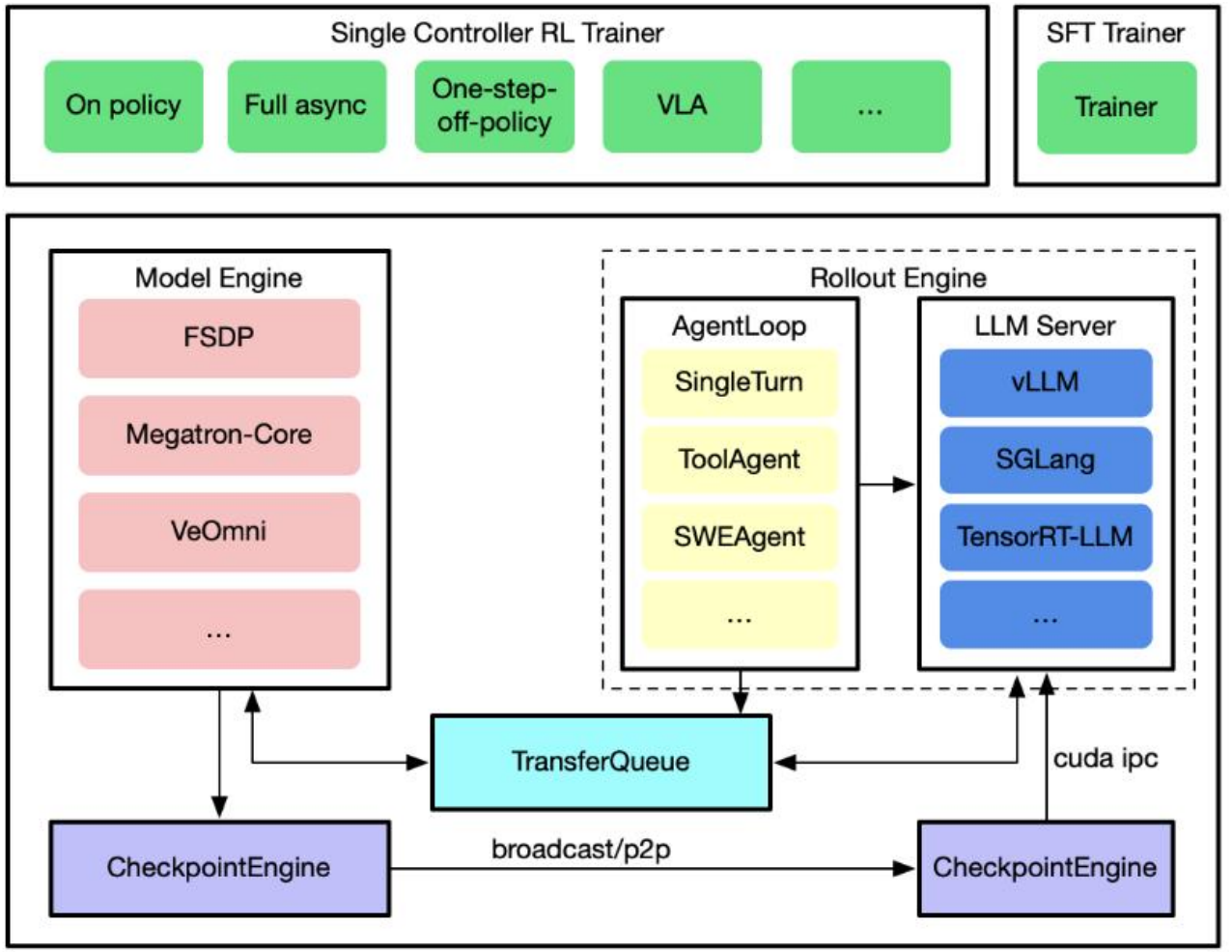
- LoRA keeps base model weights frozen and updates only low-rank adapters.
- Prior work[3] shows LoRA can mat full-parameter fine-tuning performance.



[3] LoRA Without Regret (John Schulman and Thinking Machines Lab, 2025)

Why “veRL”?

- SOTA
- Well-known
- Flexible
- Community



Performance Gap with LoRA on veRL

- Deep-Distill-Qwen on math problems under same compute budget

Model	Trainable parameters	AIME 2025 (before/after)	Headroom-normalized Δ metric	GPQA Diamond (before/after)	Headroom-normalized Δ metric
DS-Distill-Qwen-1.5B *Full*	1.5B	18.75 / 25.52	8.33%	16.24 / 37.18	25.00%
DS-Distill-Qwen-7B *LoRA, r=64*	0.16B	29.06 / 37.08	11.31%	34.26 / 52.16	27.23%
DS-Distill-Qwen-32B *LoRA, r=8*	0.07B	31.77 / 45.83	20.61%	45.43 / 63.45	33.02%

- "large prior + small LoRA" is more effective and efficient than full-parameter RL on a small model.

Existing framework for MoE on veRL

- Use mbridge[4] to transfer model weights from Mcore to HF format.



- However, mbridge only support full-parm and is being deprecated.
- Megatron-Bridge[5]
  - Inherits mbridge ideas and actively maintained.
  - Supports LoRA / PEFT.
- Framework: veRL + megatron (core/bridge) + vLLM

[4] https://github.com/ISEEKYAN/mbridge
[5] https://github.com/NVIDIA-NeMo/Megatron-Bridge

Track 1: MoE LoRA in Megatron-Bridge

- Correct LoRA support under MoE
  - Refactor the pipeline (Megatron-Bridge/pull/1603)
    - Build main tasks
    - Build adapter tasks
    - Process main tasks
    - Pair adapter tasks (for parallelism context)
    - Process paired adapter tasks
    - **Merge** (need to merge lora adaptor into base model)
  - Support Fused LoRA & Canonical LoRA
  - Flexible target_modules (pull/1799)
  - Bugfix.....

Track 1: MoE LoRA in Megatron-Bridge

- Implicit bugs in MoE LoRA supporting, e.g.
  - MoE LoRA on Expert Modules (fc1/fc2) + EP Causes Hang
    - no EP, EP + only lora attention layers
    - no ETP (ETP=TP)
    - ETP != TP, hang
  - Fix: using ETP instead of TP for expert layers

```
69  + def get_adapter_attributes_from_linear(m: nn.Module, is_expert: bool =
      False) -> Tuple[bool, int, int, bool, bool]:
70          """Returns attributes from the base layer.
71
72          input_is_parallel, in_features, out_features,
        disable_sequence_parallel_comm, base_linear_is_parallel
ool, int, int, bo
90          """
91          disable_sequence_parallel_comm = not m.config.sequence_parallel
92          base_linear_is_parallel = True
93  +       if is_expert:
94  +           tp_size = parallel_state.get_expert_tensor_parallel_world_size()
95  +       else:
96  +           tp_size = parallel_state.get_tensor_model_parallel_world_size()
97          if HAVE_TE and any(isinstance(m, te_column_parallel) for
```

Track 1: MoE LoRA in Megatron-Bridge

- Different attention structure requires specific handling
  - GQA mismatch problem (pull/1627)
  - Use **m.config.kv_channels * m.config.num_attention_heads** for calculating **q_out_features** instead of just directly using **in_features**

Fix the following error:

```
  File "megatron/core/transformer/transformer_layer.py", line 455, in forward
    hidden_states, context = self._forward_attention(*args, **kwargs)
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "megatron/core/transformer/transformer_layer.py", line 529, in _forward_attention
    attention_output_with_bias = self.self_attention(
                                 ^^^^^^^^^^^^^^^^^^^^
  File "torch/nn/modules/module.py", line 1773, in _wrapped_call_impl
    return self._call_impl(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "torch/nn/modules/module.py", line 1784, in _call_impl
    return forward_call(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "megatron/core/transformer/attention.py", line 768, in forward
    qkv_output = self.get_query_key_value_tensors(
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "megatron/core/transformer/attention.py", line 1151, in get_query_key_value_tensors
    mixed_qkv, _ = self.linear_qkv(hidden_states)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "torch/nn/modules/module.py", line 1773, in _wrapped_call_impl
    return self._call_impl(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "torch/nn/modules/module.py", line 1784, in _call_impl
    return forward_call(*args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "Megatron-Bridge/src/megatron/bridge/peft/canonical_lora.py", line 88, in forward
    return linear_output + adapter_output, bias
           ~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~
RuntimeError: The size of tensor a (2560) must match the size of tensor b (1536) at non-singleton dimension 2
```

Track 2: Integrating Megatron-Bridge into veRL

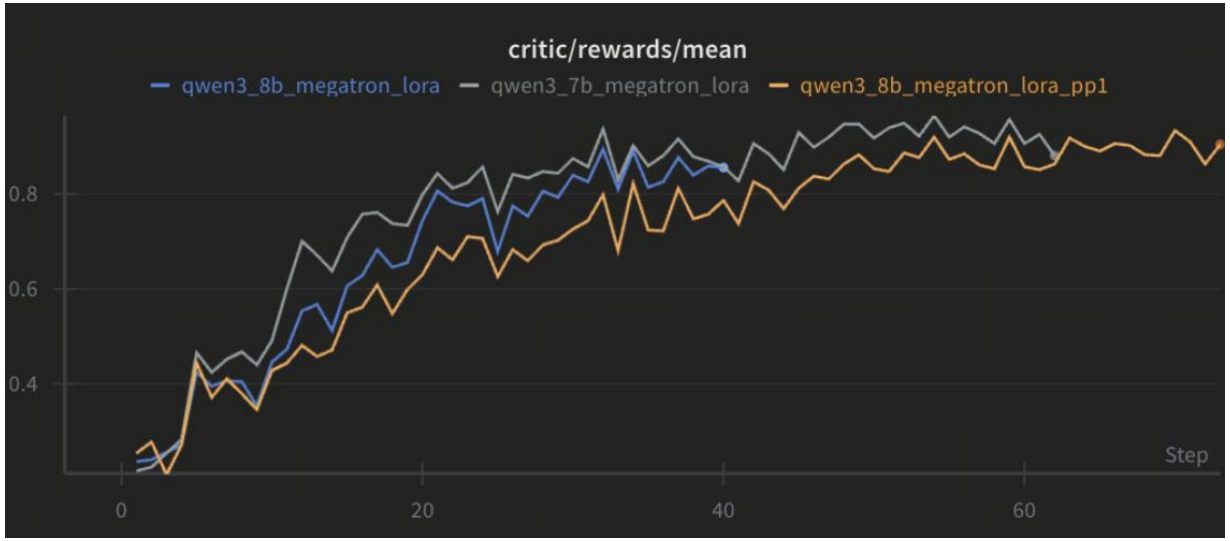
- https://github.com/volcengine/verl/pull/4063
  - How to enable megatron-bridge?

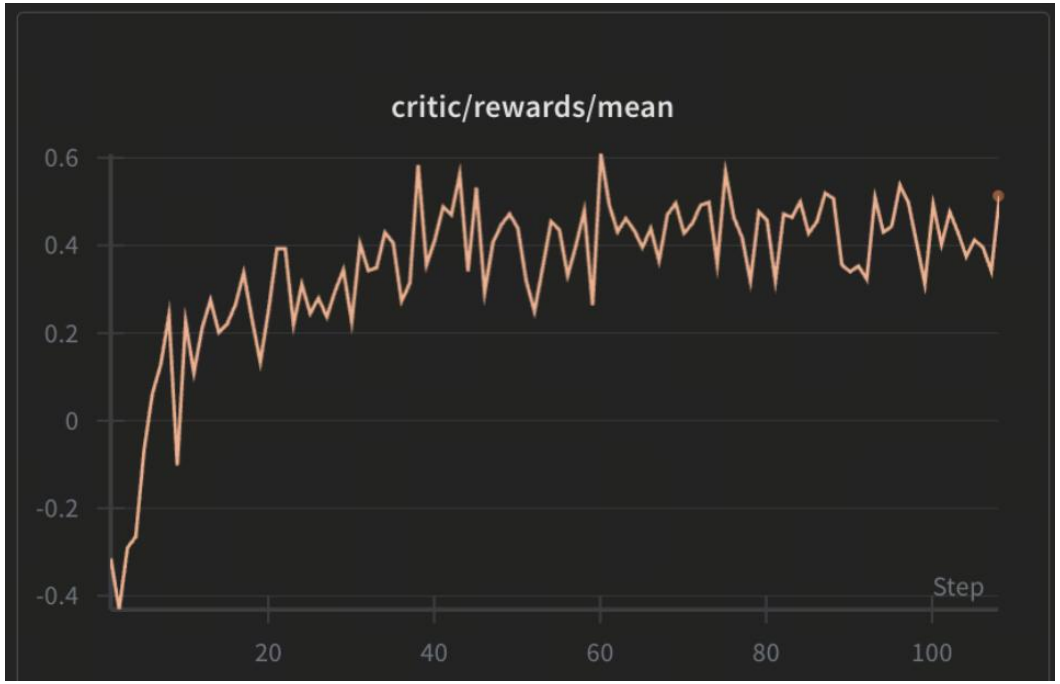
```
actor_rollout_ref.actor.megatron.use_mbridge=True
actor_rollout_ref.actor.megatron.vanilla_mbridge=False
```

  - compatible with both mbridge & Megatron-Bridge

Experiments on LoRA with veRL

- LoRA on dense / MoE models using megatron+vllm
  - examples/grpo_trainer/run_qwen2-7b_math_megatron_lora.sh
  - examples/grpo_trainer/run_qwen3moe-30b_megatron_lora.sh



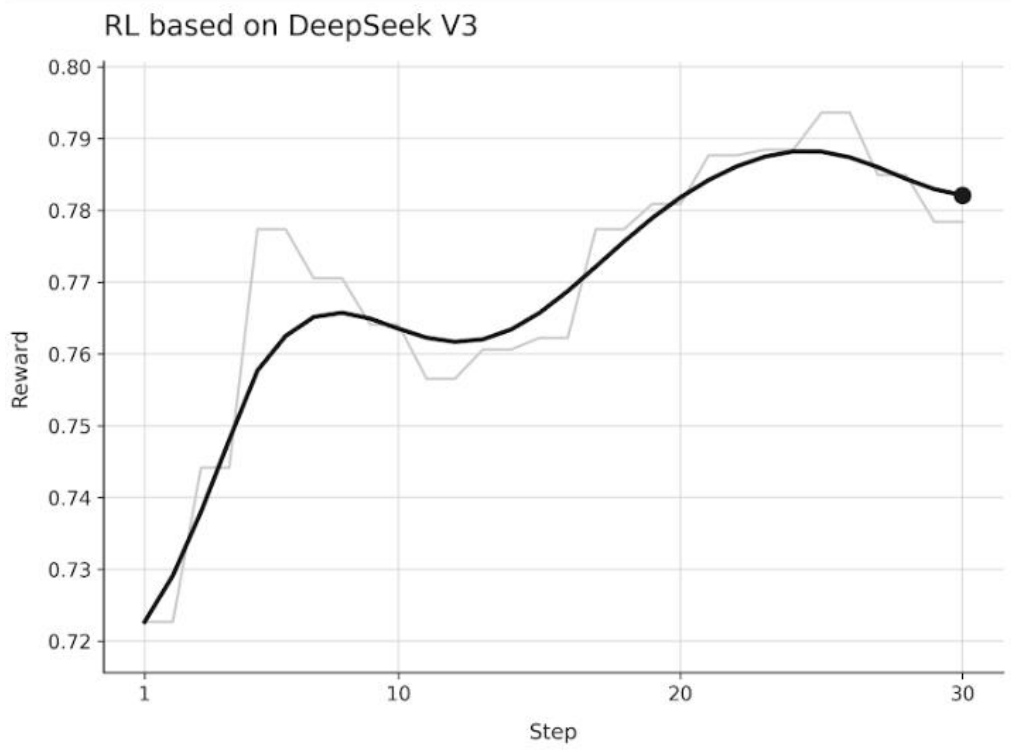


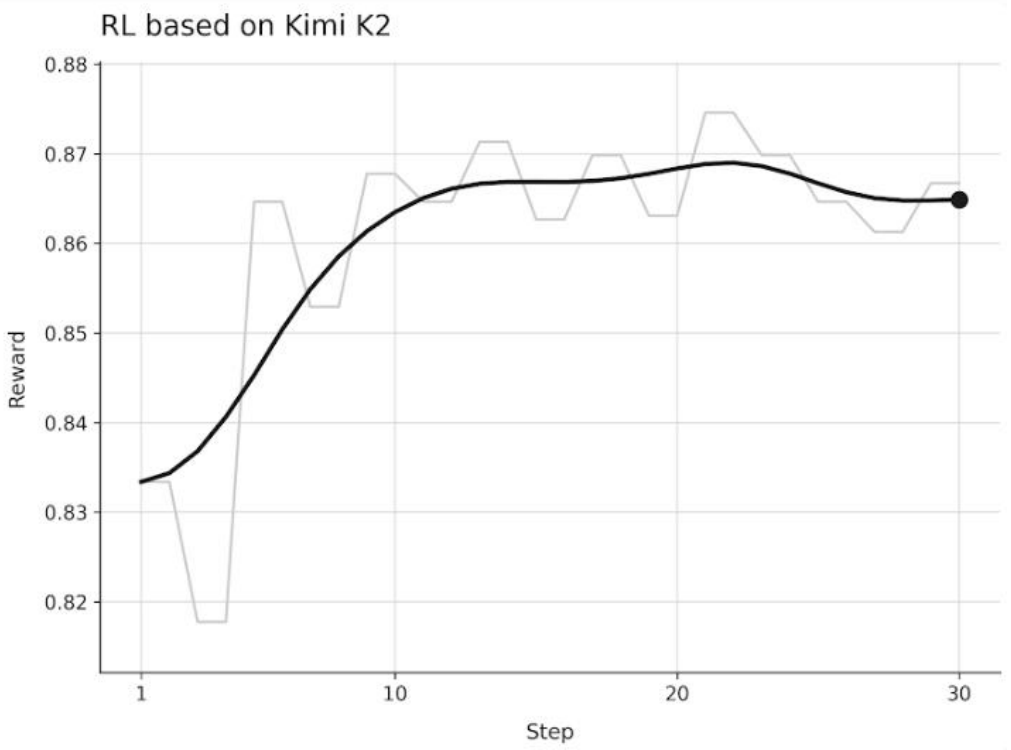
Additional Engineering for 1T MoE LoRA

- Sharded LoRA Utilities (Checkpoint engine)
  - Distributed save / load / merge / quantize
  - Reduces operational complexity at trillion scale
- Dynamic Multi-Tenant LoRA Serving
- Best parallelism strategy
- We wrap veRL and develop a hybrid framework for 1T MoE LoRA

Experiments on Large MoE LoRA

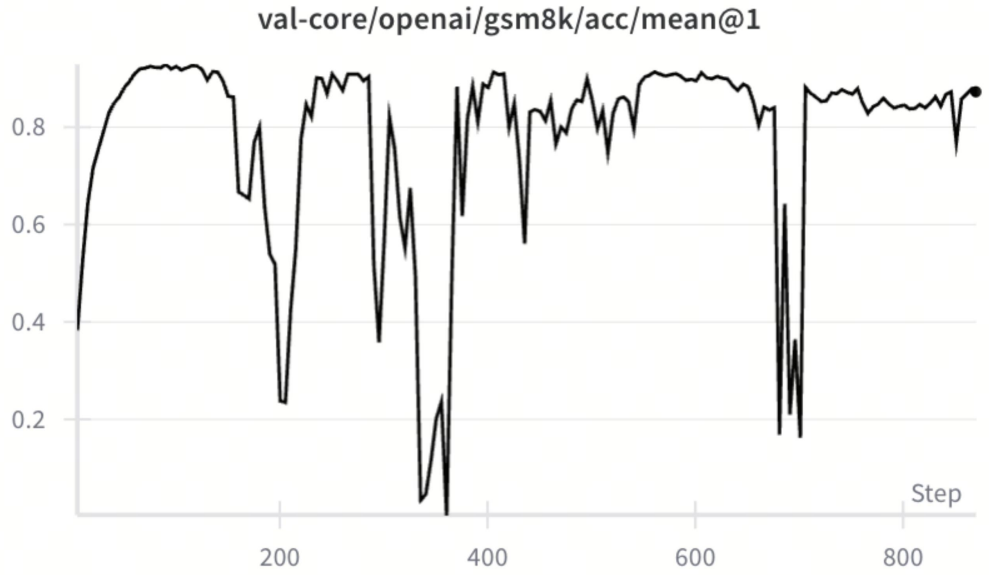
- Deepseek-V3 671B and Kimi-K2-Instruct
  - math problems, lora rank=8
  - both using 64xH800 80G





New Issues After MoE LoRA Works

- Training – Inference Mismatch
  - LoRA can introduce implementation-level mismatch on its own.
  - LoRA changes the geometry of policy updates
  - LoRA might couple into MoE router in unpredictable ways
- A LoRA RL Training Collapse Example Caused by TIM



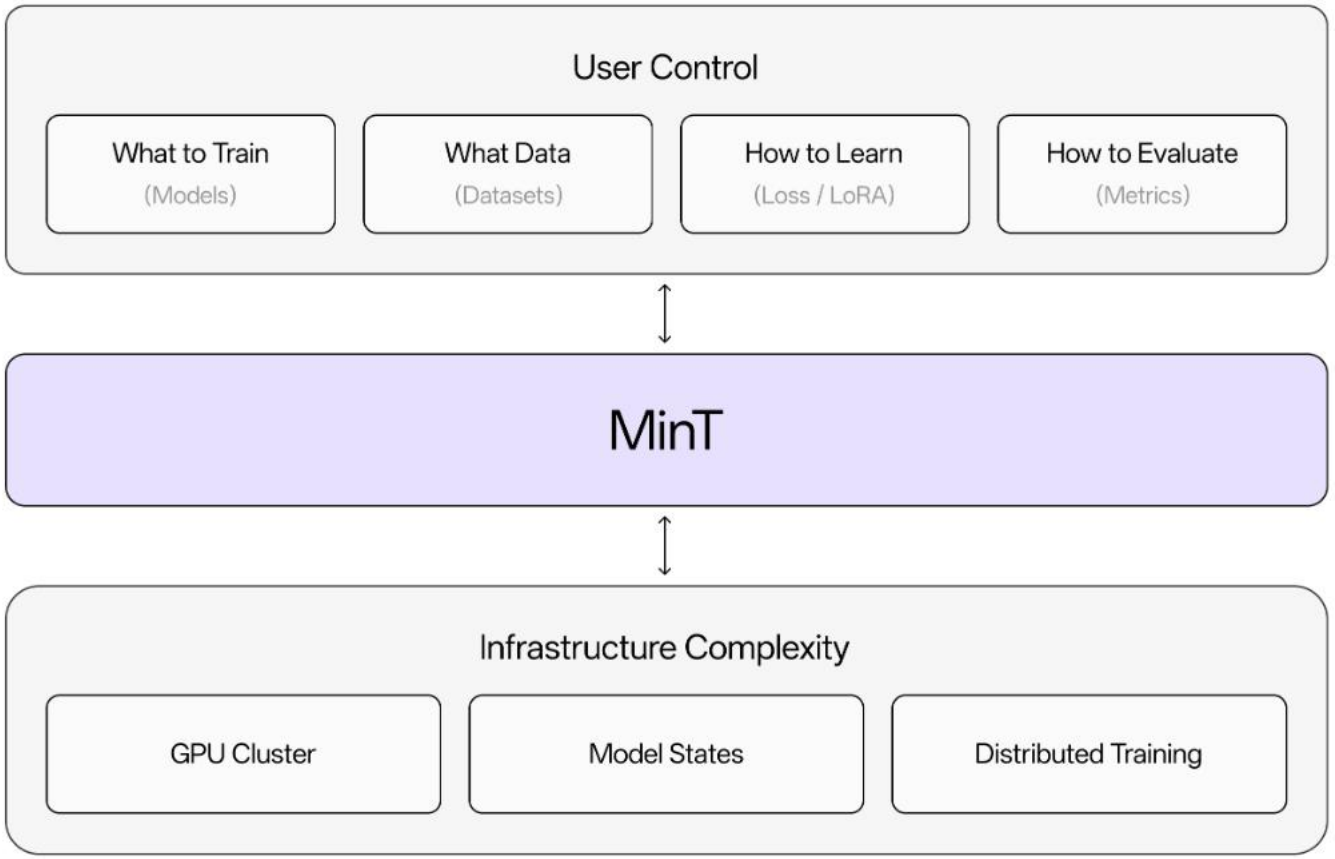
Further Optimization for LoRA RL

- Runtime LoRA Enable/Disable
  - Base model weights are frozen, LoRA on: actor, LoRA off: reference
  - Might saves memory by sharing base weights

- From LoRA Merge to LoRA adpator
  - Current: merge LoRA into base weights before export
  - Target: only synchronize LoRA adapters
    - vLLM support lora on fused moe recently
    - megatron-bridge support individually export lora recently

Make 1T MoE LoRA available for everyone

- MinT —— Mind Lab Toolkit



Mind Lab MinT 小助手

扫一扫上面的二维码图案，加我为朋友。

Thanks